

GILLETTE NORTHEAST WYOMING, WY, USA

WMO: 726650

Lat: 44.339N

Lon: 105.542W

Elev: 1327

StdP: 86.37

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 94023

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-22.4	-18.9	-26.2	0.4	-20.2	-22.7	0.6	-18.5	15.2	4.3	13.8	2.9	3.0	340	0.689

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.4	35.0	16.6	33.1	16.2	31.5	16.1	19.1	28.9	18.3	27.9	17.6	27.3	5.0	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.2	13.6	21.5	15.2	12.7	21.4	14.1	11.8	21.3	59.7	28.6	57.0	27.9	54.4	27.1	21.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.0	11.7	10.5	DB	-26.7	37.9	4.0	1.8	-29.6	39.2	-31.9	40.2	-34.2	41.2	-37.1	42.5	(4)
(5)			WB	-26.9	20.6	4.0	0.7	-29.8	21.1	-32.1	21.5	-34.3	21.8	-37.2	22.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	8.0	-3.3	-3.4	2.1	6.1	11.0	17.2	22.4	21.3	15.7	7.8	1.6	-3.3	(6)
(7)		DBStd	10.73	6.57	7.34	6.65	5.35	4.84	4.39	3.44	3.71	5.28	5.51	6.64	6.76	(7)
(8)		HDD10.0	2013	411	375	252	138	47	3	0	0	13	106	257	411	(8)
(9)		HDD18.3	4082	669	608	503	367	230	72	7	17	109	328	502	669	(9)
(10)		CDD10.0	1284	0	1	7	21	78	218	383	349	184	37	6	0	(10)
(11)		CDD18.3	313	0	0	0	0	3	38	132	107	31	1	0	0	(11)
(12)		CDH23.3	4573	0	0	2	14	88	611	1808	1468	547	34	1	0	(12)
(13)		CDH26.7	2089	0	0	0	1	18	249	905	693	217	6	0	0	(13)
(14)	Wind	WSAvg	4.7	5.0	5.0	5.1	5.1	4.8	4.3	4.0	4.1	4.3	4.7	5.0	5.0	(14)
(15)	Precipitation	PrecAvg	419	15	16	19	48	76	83	38	34	33	30	17	15	(15)
(16)		PrecMax	566	34	39	41	101	281	254	87	101	69	78	54	32	(16)
(17)		PrecMin	304	2	2	5	8	4	3	12	1	1	0	6	2	(17)
(18)		PrecStd	65	8	9	11	27	58	54	23	30	20	21	10	7	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.6	16.0	22.0	25.4	28.9	35.9	37.9	36.6	33.8	27.3	21.3	13.9	(19)
(20)			MCWB	4.7	6.7	9.0	11.3	14.6	16.4	17.3	17.1	15.4	12.3	8.7	6.0	(20)
(21)		2%	DB	9.3	12.2	18.0	22.3	25.9	32.1	35.8	34.3	31.8	23.5	17.2	10.8	(21)
(22)			MCWB	3.3	4.9	7.6	10.4	13.9	16.3	17.0	16.0	14.8	10.7	7.1	4.2	(22)
(23)		5%	DB	7.5	8.8	15.3	18.9	23.1	29.4	33.7	32.7	29.2	20.6	14.1	8.0	(23)
(24)			MCWB	2.3	3.3	6.5	9.2	13.3	16.3	16.7	15.9	13.9	9.8	5.7	2.5	(24)
(25)		10%	DB	5.8	6.2	12.3	16.1	20.8	27.1	32.0	31.0	26.8	17.4	11.4	6.2	(25)
(26)			MCWB	1.4	1.6	5.2	7.8	12.1	15.9	16.8	15.8	13.3	8.7	4.6	1.5	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.4	7.1	9.9	12.5	16.7	19.9	20.3	19.6	18.3	13.4	9.1	6.1	(27)
(28)			MCD B	11.6	15.1	20.3	23.2	24.8	27.8	30.4	31.0	27.2	23.8	19.7	13.3	(28)
(29)		2%	WB	3.7	5.1	8.1	10.8	15.2	18.5	19.3	18.5	16.2	11.6	7.5	4.3	(29)
(30)			MCD B	8.8	11.5	17.3	19.7	23.4	27.3	29.5	28.4	26.9	21.0	16.6	9.8	(30)
(31)		5%	WB	2.6	3.4	6.8	9.5	13.9	17.4	18.6	17.7	15.1	10.4	6.2	2.9	(31)
(32)			MCD B	7.1	8.4	14.6	18.0	21.7	26.7	28.4	27.5	25.9	19.1	13.7	7.6	(32)
(33)		10%	WB	1.4	1.8	5.5	8.3	12.6	16.3	17.8	16.9	14.0	9.1	4.8	1.6	(33)
(34)			MCD B	5.4	5.8	11.8	15.3	19.6	25.5	27.7	26.9	24.7	16.7	11.1	5.9	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	10.8	11.2	12.6	13.1	13.0	15.0	16.4	16.4	15.5	13.2	12.3	11.0	(35)
(36)			MCDBR	11.9	13.6	17.1	18.3	17.0	18.8	18.8	19.2	18.3	16.3	12.9	(36)	
(37)			MCWBR	7.6	7.9	8.8	8.9	7.7	7.4	6.0	6.2	7.1	8.5	8.9	8.0	(37)
(38)		5% WB	MCDBR	11.7	12.9	16.0	16.6	15.7	16.9	16.5	16.3	16.9	16.6	15.5	12.3	(38)
(39)			MCWBR	7.7	8.0	8.5	8.6	7.7	7.6	6.0	6.0	6.6	8.2	8.7	7.9	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.245	0.252	0.270	0.295	0.320	0.328	0.350	0.350	0.312	0.281	0.257	0.243	(40)	
(41)		taud	2.581	2.572	2.567	2.486	2.477	2.474	2.435	2.417	2.515	2.581	2.565	2.574	(41)	
(42)		Ebn at Noon	916	965	981	972	950	939	912	902	917	913	888	883	(42)	
(43)		Edh at Noon	62	75	86	102	106	107	110	108	90	73	62	56	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.84	2.67	3.94	5.02	6.04	7.13	7.29	6.26	4.81	3.13	2.05	1.50	(44)	
(45)		RadStd	0.11	0.17	0.34	0.31	0.58	0.52	0.27	0.20	0.33	0.27	0.18	0.12	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (9 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page